

Understanding and Evaluating Routing Protocols in Computer Networks

Course: Computer Networks

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Objective

To understand the principles of routing, distinguish between different types of routing protocols, and analyze their performance and use in real-world network environments.

1. Introduction to Routing and Routing Protocols

- Routing in computer networks is the process of selecting a path for traffic in a network, or between or across multiple networks. It is a key function of the network layer (Layer 3) in the OSI model.
- Routing protocols are used to facilitate the exchange of routing information between routers, enabling them to build and maintain routing tables.
- Static routing is manually configured and remains constant unless changed by an administrator. Dynamic routing uses algorithms and routing protocols to adjust paths automatically based on network conditions.

2. Classification of Routing Protocols

Routing protocols are essential for determining the best paths for data packets to travel across a network. They can be broadly classified into three main categories: Distance Vector, Link State, and Hybrid protocols. Each type has distinct characteristics, working principles, and use cases.

1. Distance Vector Routing Protocols

Distance Vector (DV) protocols determine the best path to a destination based on distance (usually hop count) and direction (next-hop router). These protocols periodically share their entire routing tables with neighboring routers.

Key Characteristics:

Periodic Updates: Routers exchange routing tables at regular intervals.

Hop Count as Metric: The primary metric is the number of hops to reach a destination.

Slow Convergence: Updates propagate slowly, leading to longer convergence times.

Prone to Routing Loops: Uses mechanisms like Split Horizon, Poison Reverse, and Hold-Down Timers to prevent loops.

Example: Routing Information Protocol (RIP)

Working Principle:

- RIP uses **hop count** as its metric (max 15 hops).
- Updates are broadcast every **30 seconds**.
- If a route is not refreshed within **180 seconds**, it is marked as invalid.
- Uses the **Bellman-Ford algorithm** to compute paths.

Limitations:

- Not suitable for large networks due to the **15-hop limit**.
- High overhead due to frequent updates.
- Slow convergence co-working Principle

2. Link State Routing Protocols

Link State (LS) protocols maintain a complete topological map of the network. Each router independently calculates the shortest path using Dijkstra's algorithm.

Key Characteristics:

- Topology Database: Each router maintains a Link State Database (LSDB).

- **Event-Driven Updates:** Updates are sent only when a change occurs (e.g., link failure).
- **Fast Convergence:** Reacts quickly to network changes.
- **Higher Resource Usage:** Requires more CPU and memory due to complex calculations.

Example: Open Shortest Path First (OSPF)

Working Principle:

- Divides networks into areas to improve scalability.
- Uses Dijkstra's algorithm to compute the shortest path.
- Metric is based on bandwidth ($\text{cost} = 10^8 / \text{BW in bps}$).
- Routers exchange Link State Advertisements (LSAs) to update the topology.

Advantages:

- Supports hierarchical routing (large networks).
- Faster convergence than RIP.
- Supports VLSM (Variable-Length Subnet Masking) and CIDR.

3. Hybrid Routing Protocols

Hybrid protocols combine features of Distance Vector and Link State protocols to improve efficiency and scalability.

Key Characteristics:

- **Balanced Approach:** Uses distance-vector metrics but exchanges topology information like link-state protocols.
- **Faster Convergence:** More efficient than pure DV protocols.
- **Reduced Overhead:** Sends updates only when necessary.

Example: Enhanced Interior Gateway Routing Protocol (EIGRP)

Working Principle:

- Uses Diffusing Update Algorithm (DUAL) for loop-free paths.
- Metric is based on bandwidth, delay, reliability, load, and MTU.
- Maintains neighbor tables and topology tables.
- Sends partial updates (only when changes occur).

Advantages:

- Faster convergence than RIP and OSPF in some cases.
- Supports unequal-cost load balancing.
- Cisco proprietary (but now partially documented as an IETF standard).

3. Comparative Analysis

The following table compares RIP, OSPF, and EIGRP based on several key parameters.

Protocol	Routing Type	Convergence Speed	Metric Used	Scalability	Protocol Overhead	Use Cases & Environments
RIP	Distance Vector	Slow	Hop Count	Low	Low	Small networks
OSPF	Link State	Fast	Cost (Bandwidth)	High	High	Large enterprise networks
EIGRP	Hybrid	Faster than RIP, close to OSPF	Composite metric (Bandwidth, Delay, etc.)	High	Medium	Medium to large networks